BANGALORE INSTITUTE OF TECHNOLOGY

An Autonomous Institution under Visvesvaraya Technological University, Belagavi
K. R. Road, V. V. Puram, Bengaluru - 560004

DEPARTMENT OF CSE/ISE/AI&ML/CSE(DS)/CSE(ICB)

MODULE-WISE QUESTION BANK

Academic Year :	2026-27	Batch :	2025
Course Code :	1BCS304	Course Title :	Operating Systems
Credits:	3	Total Hours of Pedagogy:	42:0:0:48=90
Total Contact Hours:	42		

Guidelines:

- Questions shall be mapped to the appropriate Course Outcome (CO).
- Bloom's cognitive level shall be indicated as L1 to L6, wherever applicable.
- Question category may be indicated as Conceptual / Numerical / Application / Analysis / Design / Case Study / Programming / Other, as applicable.
- The question bank shall provide adequate coverage of all modules and relevant COs.

Bloom's Taxonomy Reference

L1	L2	L3	L4	L5	L6
Remember	Understand	Apply	Analyse	Evaluate	Create

MODULE - 1:

Q. No.	Question	CO	RBL
1	Explain in detail the various functions performed by an operating system. Discuss the OS as a resource manager.	1	L2
2	With a neat diagram, explain computer system organization and describe the interrupt-driven I/O cycle.	1	L2
3	Explain the storage-device hierarchy with a labelled diagram. Discuss the trade-offs between speed, cost, and volatility across levels.	1	L4
4	Differentiate between single processor systems, multiprocessor systems, and clustered systems with diagrams. Discuss the pros and cons of each.	1	L4
5	Explain symmetric and asymmetric multiprocessing architectures with diagrams.	1	L2
6	Describe the dual-mode operation of an operating system. Explain how the mode bit helps in protecting the OS from user processes, with a neat diagram showing transition between user mode and kernel mode.	1	L2
7	Explain the following operating-system operations in detail: (a) Multiprogramming (b) Time sharing (c) Real time (d) Network (e) Distributed	1	L2
8	Discuss the responsibilities of the operating system under: (a) Process management (b) Memory management (c) Storage management. Explain with relevant activities performed under each.	1, 2	L2
9	Describe the User–Operating System interface. Compare CLI and GUI-based interfaces with examples of OS that support each.	1	L4
10	Define a system call? Explain, with a diagram, how a system call is executed (e.g., the read() system call), including the role of the system call interface and system-call table.	1	L2

Q. No.	Question	CO	RBL
11	Explain the different types of system calls with examples for each type (process control, file management, device management, information maintenance, communications, protection).	1	L2
12	A programmer writes fork() followed by exec() in a C program on a UNIX-like system. Trace the sequence of system calls invoked, explain the purpose of each, and describe what happens to the parent and child processes.	2	L4
13	Illustrate, with an example, how parameters are passed to the operating system during a system call. Explain the three methods used for parameter passing.	1	L3
14	Define system programs? Explain the different categories of system programs with examples of each.	1	L2
15	Discuss the issues involved in operating system design and implementation. Explain the concepts of mechanism and policy with examples.	1	L4
16	Describe simple structure (MS-DOS), layered structure, and microkernel structure of operating systems with diagrams, discussing advantages and disadvantages of each.	1	L4
17	Explain the concept of loadable kernel modules and how they provide flexibility over a purely layered or monolithic structure.	1	L2
18	Compare monolithic, layered, microkernel, and modular OS structures based on performance, ease of extension, and reliability.	1	L4
19	Explain the role of an operating system as a resource allocator and control program, giving examples of conflicting requests it must resolve.	1	L3
20	A team is designing an OS for a resource-constrained IoT device that needs high reliability but minimal hardware access exposure to third-party services. Recommend a suitable OS structure (layered/microkernel/monolithic) and justify your choice with reference to advantages and drawbacks.	1	L5
21	A vendor wants to add a new file system driver to a running OS kernel without rebooting the system. Explain how loadable kernel modules make this possible, and compare this approach with a purely layered structure where such changes are harder.	1	L4
22	Linux is often described as a hybrid, monolithic-based system with modular components. Justify this description by explaining which structural principles (from monolithic, layered, microkernel, and modular approaches) it combines, and why.	1	L5

Faculty In-charge

Course Coordinator **Module Coordinator** **IQAC** **Program Coordinator**